How much the atmosphere moves

What dynamic range must the instrument and the inversion survive?

`Notebook companion: Unit-03-variability.ipynb`

1 The question

Unit 2 described the atmosphere as though it had one shape. It does not. At a fixed altitude the density varies by a factor of about five over a solar cycle, and by another factor of about five within a day during a geomagnetic storm. The composition changes as well as the amount.

That variability is not a small nuisance around a design point. It is the reason the instrument exists. An atmosphere that behaved predictably would not need measuring, because a model fitted to its average would serve. What makes the measurement worth flying is that the atmosphere departs from its average by large factors, at short notice, and in ways the models do not reproduce. The design case is therefore a disturbed atmosphere rather than a typical one. This unit puts numbers on that variability so the choice can be defended rather than asserted.

2 Physical picture

Two inputs drive the thermosphere, on very different timescales.

Solar extreme-ultraviolet irradiance heats the upper atmosphere directly, and it varies with the 27-day solar rotation and with the eleven-year cycle. It is indexed by F10.7, the radio flux at 10.7 cm, which is measured daily from the ground and correlates well enough with the unmeasured extreme ultraviolet to serve as a proxy.

Geomagnetic activity heats it too, through currents driven by the solar wind that deposit energy at high latitudes over hours rather than days. It is indexed by ap. That heating is confined in latitude and brief in time, and the disturbance it creates then travels towards lower latitudes over the following hours.

Both kinds of heating raise the temperature. A higher temperature gives a larger scale height. A larger scale height raises the density at every altitude above the height where the heating occurred. All of that is a change in the amount of air. Storm heating also drives *upwelling*, which lifts molecular-rich air from below into a region that had become atomic. That is a change in composition, and a model that scales a whole density profile by one factor cannot represent it. Section 4b measures the difference between the two.

One further variation matters because of where we measure. Occultations happen only at the terminator (Unit 5), so every profile is a dawn or a dusk profile, and those are not the same atmosphere. Dusk is the warmer, more molecular one.

3 Derivation: why the range widens with altitude

Density falls as $\exp\left(-\int dz/H\right)$ with $H = kT/mg$. Heating changes T, and the effect on density at a given altitude compounds with the number of scale heights between the heating and the observation. Expanding to first order,

$$\frac{\delta n}{n} \simeq \frac{z - z_{\mathrm{ref}}}{H} \frac{\delta T}{T}.$$

The consequence is that the required dynamic range is a function of altitude. A severe storm raises the exospheric temperature from 838 to 1165 K, a fractional change of 0.39, and with a scale height of 44 km for atomic oxygen the expansion predicts the density ratio at 400 km to be 2.75 times its value at 200 km. The computed ratio is 2.64. At 300 km the expansion predicts 1.88 against 1.54 computed, so the argument reproduces the trend and overstates its size. That is what a one-term expansion should do when the fractional temperature change is as large as 0.39.

Treat the expansion as a scaling argument only. It shows that the range of densities the instrument must cover grows with altitude, and it shows why. Section 4a gives the numbers.

4 Worked numbers

4a. The range of densities the instrument must cover

Case	**200** km	**300** km	**400** km
Solar minimum, quiet	0.62	0.33	0.17
Moderate, quiet	1.00	1.00	1.00
Solar maximum, quiet	1.70	2.44	3.46
Moderate, ap = 100	1.26	1.52	1.92
Moderate, ap = 400	1.78	2.75	4.71
Solar maximum, ap = 400	2.90	5.66	11.86
Extreme to extreme	**4.7**	**17.1**	**69.1**

Atomic oxygen density relative to a moderate quiet atmosphere at the same altitude.

The bottom row is the range the instrument must cover at each altitude: a factor of 4.7 at 200 km and a factor of 69 at 400 km. Compare that with the usable optical-depth window of Unit 6, which spans τ from 0.2 to 4 and therefore covers a factor of twenty in column. A single channel cannot cover a factor of 69, so at one extreme of the range or the other its optical depth falls outside the usable window. That is the quantitative reason for a set of channels whose altitude bands are staggered rather than one well-chosen filter, and it is why the set has to be evaluated against disturbed atmospheres and not only quiet ones.

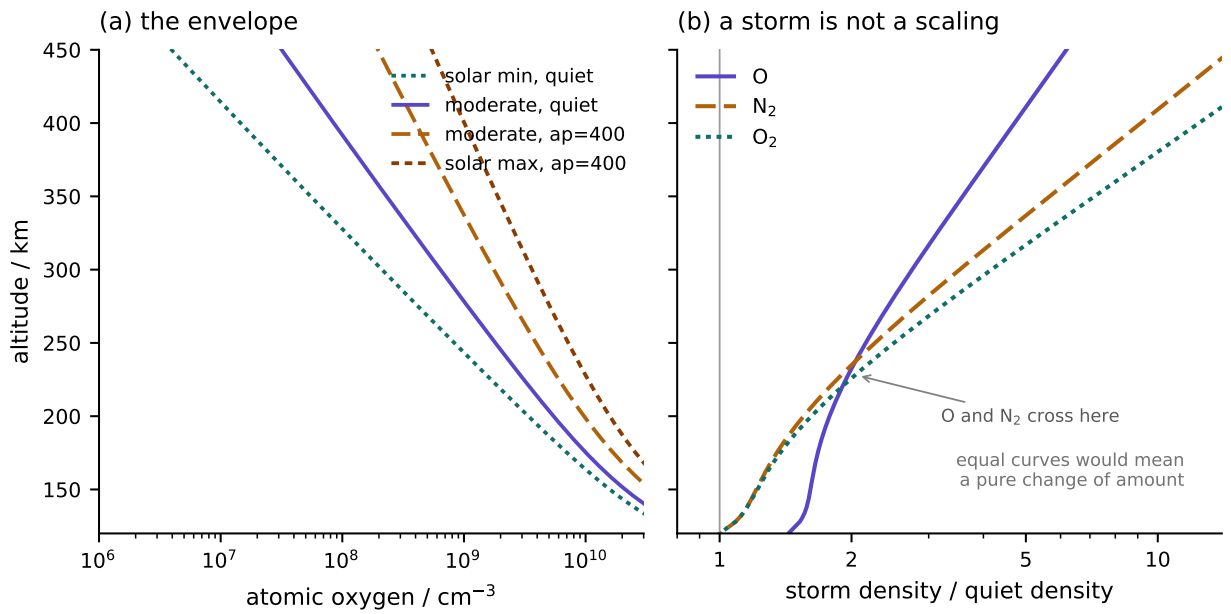


Figure 1: (a) Atomic oxygen across four of the six cases. The spread widens upward, because a temperature change compounds over the scale heights between the heating and the observation. (b) Storm density divided by quiet density for each species. If a storm were a change of amount, the three curves would coincide. Generated by sections A and C of the notebook.

4b. A storm is not a change of amount

Altitude	O	N_2	O_2	O/N_2 quiet	O/N_2 storm
150 km	1.61	1.21	1.22	0.83	1.10
200 km	1.78	1.55	1.62	2.04	2.34
250 km	2.15	2.26	2.50	4.55	4.34
300 km	2.75	3.53	4.17	9.91	7.72
400 km	4.71	9.16	12.42	45.04	23.16

Storm density divided by quiet density, at ap = 400. Equal ratios across the row, and an unchanged O/N_2, would mean the storm had only changed the amount of air.

The ratios along each row are not equal, and the composition ratio changes by a factor of two at 400 km. The change also reverses sign with altitude. Below about 250 km a storm makes the atmosphere *more* atomic, because the heaviest species have the smallest scale heights and so gain the least density from a temperature rise. Above 250 km a storm makes the atmosphere *more* molecular, because upwelling carries molecular air into a region that was atomic, and that transport changes the composition more than the scale-height effect does. A retrieval that assumes a fixed composition, or that fits composition as a single scale factor, is therefore wrong in opposite directions on either side of that crossover.

Both terminators are sampled, and they differ as well. At 200 km, dawn has an O/N_2 ratio of 2.04 and an exospheric temperature of 838 K; dusk has 1.27 and 1137 K. Dusk is hotter and more molecular. Half of our profiles are of each, and Unit 1 notes that the closest prior measurement

found NRLMSIS wrong by 26 per cent specifically at dawn.

4c. Structure a smooth model cannot follow

Everything above leaves the atmosphere smooth, so a smooth model can in principle follow it. The project therefore tests against three cases that impose vertical structure a two-parameter temperature profile provably cannot fit. Each magnitude is a named constant in `truth.py` with its justification, and each is deliberately at the strong end of the observed range.

Case	Altitude	O	N_2	O_2	ΔT
Heating layer, 200 K at 180 km	150 km	0.83	0.70	0.53	+114 K
	250 km	1.27	1.35	1.41	+9 K
Composition depletion, O by 2.5	150 km	0.78	1.18	1.18	0
	200 km	0.40	1.50	1.50	0
Turbopause raised to 160 km	130 km	0.17	1.23	2.45	0
	150 km	0.12	1.46	4.14	0

Density relative to the quiet reference. The three cases violate three different assumptions: the shape of the temperature profile, the constancy of composition, and diffusive equilibrium itself.

The third case is the most severe and the most localised. Raising the turbopause to 160 km reduces atomic oxygen at 150 km by a factor of eight, raises molecular oxygen by a factor of four, and changes nothing at all above 200 km. It violates the assumption of diffusive equilibrium that the retrieval makes, rather than only moving the parameters the retrieval fits, and that is why it is in the set.

4d. What the disturbance does to the instrument

The optical-depth window is a property of the hardware and does not move. The atmosphere does, so the altitudes at which each channel sits inside that window move with it.

Channel	Quiet	ap = 100	ap = 400	Bottom rises
Zirconium	166–321 km	172–350 km	183–408 km	+17 km
Aluminium	227–371 km	241–408 km	268–480 km	+41 km
Indium	168–310 km	171–335 km	176–385 km	+8 km
FUV 135–155	141–194 km	142–199 km	144–209 km	+3 km
FUV 155–175	114–155 km	114–157 km	114–161 km	0 km

Every band in the table moves upward, and the bands move by different amounts. The floor of the aluminium channel rises 41 km, while the floor of the 155–175 nm channel does not move at

all. The reason is which species each passband absorbs against. The aluminium passband overlaps the absorption of atomic oxygen and nitrogen, and the scale heights of those two species respond strongly to heating. The 155–175 nm passband overlaps the absorption of molecular oxygen, which is the heaviest of the three, has the smallest scale height, and is therefore displaced least by a temperature rise.

That has a design consequence worth stating plainly. In a disturbed atmosphere the lower part of each extreme-ultraviolet channel's band moves above 175 km, so the only channels still measuring below that altitude are the far-ultraviolet ones. Keeping the far-ultraviolet channels is what preserves low-altitude coverage during a storm, and that is a different argument for keeping them than the species argument in Unit 10.

5 What the computation shows

The required dynamic range is a function of altitude. A factor of 4.7 at 200 km, 17 at 300 and 69 at 400. The usable optical-depth window spans a factor of twenty in column, so one channel cannot cover the factor of 69 required at 400 km, and any single-channel design is being validated on the easy part of the problem.

A storm changes what the atmosphere is made of, and the sign of the change flips with altitude. Below 250 km a storm makes the air more atomic; above it, more molecular. A retrieval that treats composition as one number is wrong in opposite directions across that boundary.

The channel altitude bands move upward by different amounts under disturbance. The extreme-ultraviolet floors rise by 17 to 41 km, while the far-ultraviolet floors rise by 3 km and by nothing. During a storm the far-ultraviolet channels are the only ones still measuring at the bottom of the range.

6 What we decided

The design case is a disturbed atmosphere. Every performance figure in this series is reported for both a quiet and a severely disturbed case, and where only one is quoted it is the disturbed one. The justification is section 4a: the quiet case is a factor of five easier at 400 km, and the states in which the measurement is most valuable are the states in which it is hardest.

The retrieval is tested against structure it cannot represent, not only against atmospheres it can. The three cases of section 4c are that test, their magnitudes are named constants that can be challenged, and Unit 8 reports what they do.

The far-ultraviolet channels are justified twice: once for the species they add, and once for the low-altitude coverage they retain when the atmosphere is disturbed. The second argument is section 4d, and it was not part of the earlier channel-plan reasoning.

7 What we do not claim

Not that the model represents storms accurately. Every number in this unit comes from NRLMSIS driven by ap. An empirical model driven by a three-hourly planetary index produces a smooth, averaged profile, and a real storm is neither smooth nor an average. The real atmosphere

has structure the model does not, which is precisely why section 4c exists. Read the Tier 1 storm cases as a lower bound on the variability rather than a description of it.

Not that ap = 400 is a worst case. It is near the top of the index, but the index saturates and the largest storms are not well represented by it. We call it severe, not extreme.

Not that the local-time comparison is complete. Dawn and dusk here are 06:00 and 18:00 at the equator at equinox. The terminator's local time varies with season and orbit, and the seasonal and hemispheric asymmetries of the thermosphere are large. This unit shows that the two terminators differ; it does not survey how much.

Not that the Tier 2 magnitudes are calibrated. They are chosen at the strong end of the observed range to find a breaking point, and `truth.py` says so. A result that survives them is robust; a result that fails them is not necessarily a failure against the real atmosphere.

8 Exercises

E3.1 The scaling argument of section 3 was tested against a storm. Test it against a solar-cycle change instead, comparing solar minimum with solar maximum at fixed ap. Does the expansion work better or worse, and which of its assumptions explains the difference?

E3.2 The zirconium channel's floor rises 17 km under disturbance and the 155–175 nm channel's does not move. Work the argument through quantitatively: compute the scale height of each absorbing species in both atmospheres and show that the difference in floor movement follows.

E3.3 Take the composition-depletion case and predict, before computing, what a retrieval assuming diffusive equilibrium would report for the exospheric temperature and for the oxygen density at 200 km. Then run `occ.monte_carlo` on it and compare. State which of your two predictions you were more confident about and whether that confidence was justified.

References

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[2] J. T. Emmert, D. P. Drob, J. M. Picone, D. E. Siskind, M. Jones Jr. et al., "NRLMSIS 2.0: a whole-atmosphere empirical model of temperature and neutral species densities," *Earth Space Sci.* **8**(3), e2020EA001321 (2021), doi:10.1029/2020EA001321.

[3] R. K. Sewell et al., "Thermospheric composition and temperature from GOES-R/SUVI solar occultations," *Space Weather* **23**, e2024SW004234 (2025); arXiv:2508.10242.

[4] M. F. Storz, B. R. Bowman, J. I. Branson, S. J. Casali and W. K. Tobiska, "High accuracy satellite drag model (HASDM)," *Adv. Space Res.* **36**(12), 2497–2505 (2005), doi:10.1016/j.asr.2004.02.020.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.